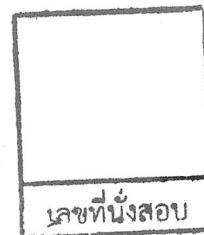


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of Semester 1

Year: 2018

Subject: 392153 Chemistry III

Section: 15-18

Date: 26 November 2018

Time: 8.00-10.00

Name: _____ ID: _____ Class: _____

Instructions:

1. The examination has 2 parts, 15 pages (including this page), 8 questions and a total score of 80 marks (40%)
 2. Write all your solutions and answers on this examination sheet.
 3. This is a closed book examination.
 4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
 5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
 6. You are not allowed to use the restroom during the exam except in case of an emergency.
 7. No documents are allowed to be taken out of the examination room.
 8. Calculators are not allowed in the examination.
 9. Electronic communication devices are NOT allowed in the examination room.
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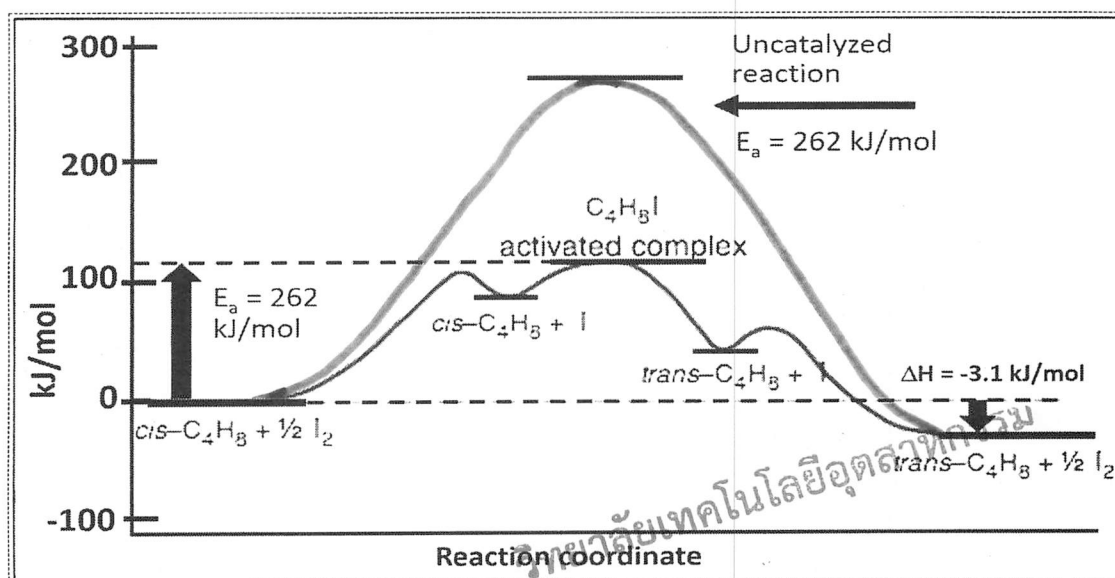
Cheating in the exam is considered an extremely serious offence
which will result in expulsion from the university.

INSTRUCTIONS: Written Response

Students will be expected to communicate your knowledge and understanding of chemical principles in a clear and logical manner. Students' steps and assumptions leading to a solution must be written in the spaces below the questions. Answers must include units where appropriate and be given to the correct number of significant figures. For questions involving calculation and explanation, full marks will NOT be given for providing only an answer.

Part 1: Total score 40 marks = 20%

1. Using the data illustrated in the graphs below to answer the following equations (10 marks)



- 1.1 Why are the reaction energy barrier and reaction pathways changed by introducing a catalyst into a chemical reaction?
- 1.2 How do they change?
- 1.3 What is the advantage of those changes?

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3. Answering about collision theory (10 marks)

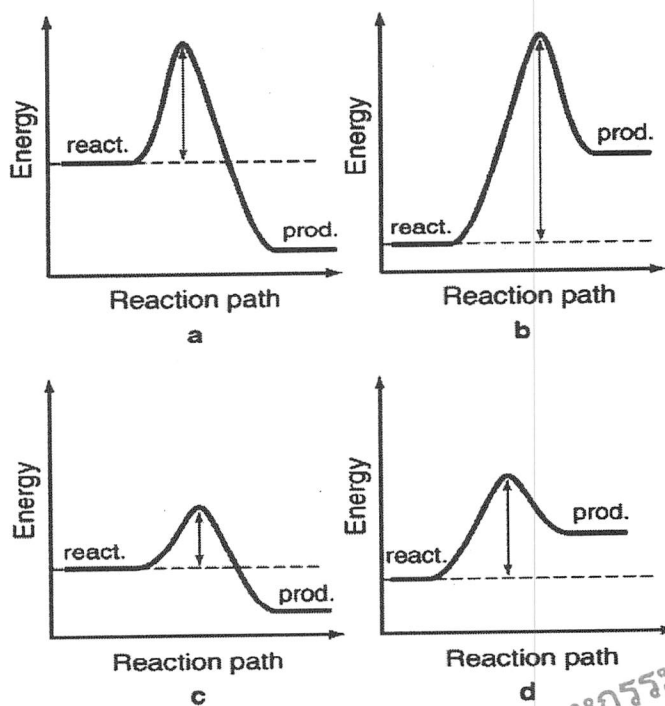
3.1 When molecules collide each other, it does not mean that reaction will occur.

What can be used to determine if a collision of molecules will be efficient?

3.2 The energy profiles for four different reactions are shown. Which reaction

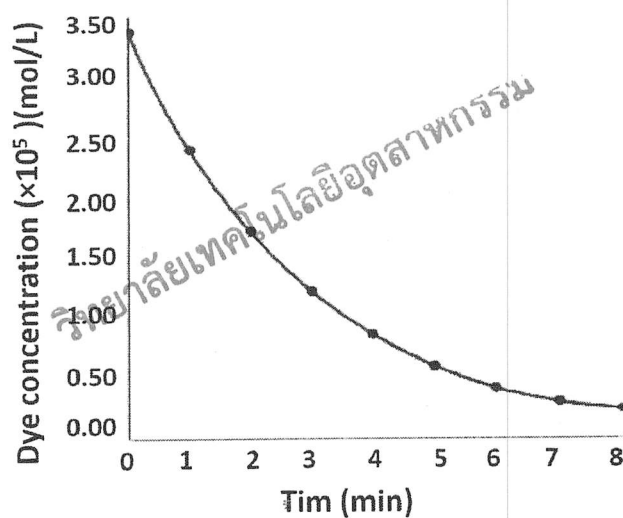
requires the most energetic collisions to reach the transition state?

Explain in terms of collision theory assumption.



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4. Data collected on the concentration of dye as a function of time are given in the graph below. (10 marks)



4.1 What is the average rate of change of the dye concentration over the first 2 min?

4.2 What is the average rate of change during the fifth minute (from $t = 4$ min to $t = 5$ min)?

4.3 Estimate the instantaneous rate at 4 min.

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Part 2: Total score 40 marks = 20%

1. Determine the rate law and calculate the rate constant for the following reaction

from the following data: $2A + 2B + C \rightarrow X + Y$ (10 marks)

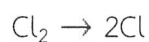
Experiment	[A] (M)	[B] (M)	[C] (M)	Initial Rate (M/s)
1	0.1	0.1	0.1	0.12
2	0.2	0.1	0.1	0.24
3	0.1	0.1	0.2	0.12
4	0.1	0.2	0.2	0.24
5	0.3	0.1	0.3	0.36

1. Determine the rate law for this reaction?
2. Calculate the rate constant (k) and give its units?
3. What is the order of reaction?

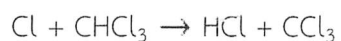
[illegible]

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2. Consider the following mechanism of carbon tetrachloride (CCl_4) (10 marks)



E_a of this reaction is 800 kJ



E_a of this reaction is 400 kJ



E_a of this reaction is 500 kJ

2.1 Write the overall reaction equation?

2.2 What are intermediates?

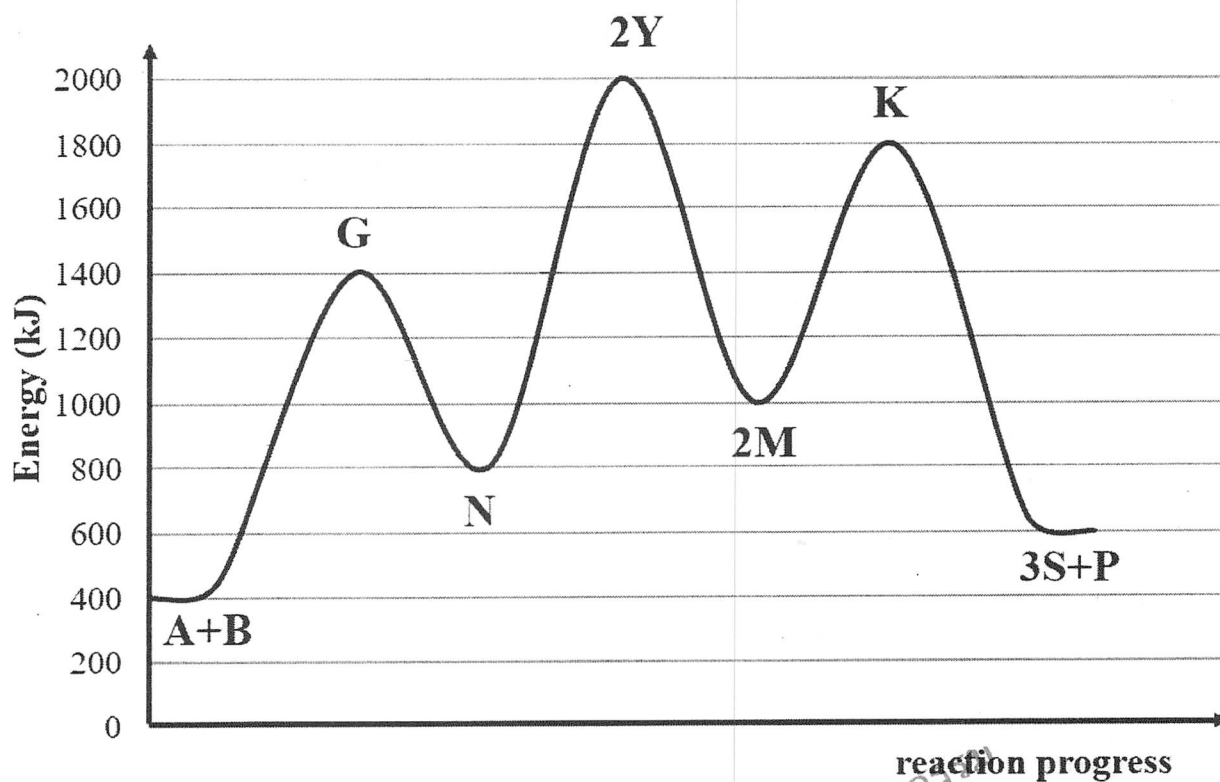
2.3 Identify any catalysts in this reaction?

2.4 Determine rate of reaction?

2.5 Which is the rate-determining step of this reaction?

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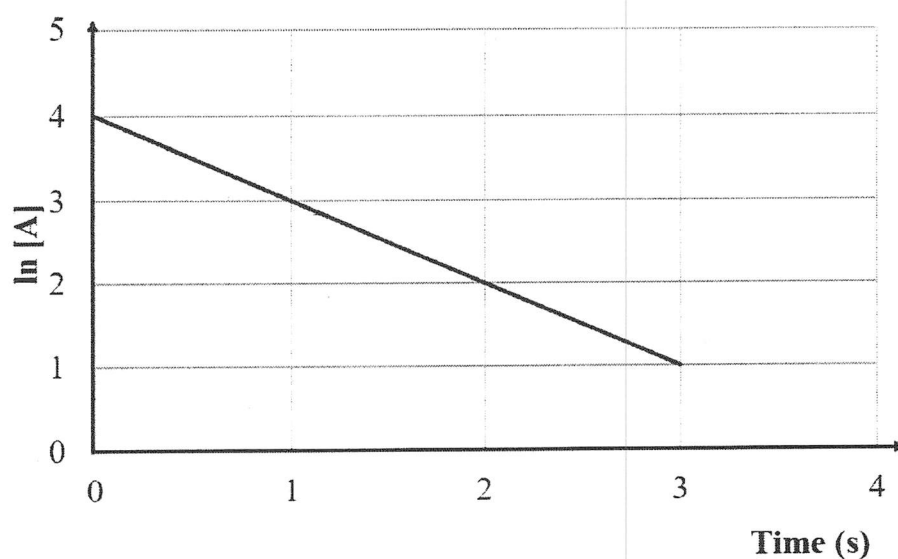
3. Consider the reaction graph below. (10 marks)



- 3.1 Write the mechanism for each reaction?
- 3.2 Write the overall reaction equation?
- 3.3 What are intermediates?
- 3.4 Calculate E_a for each reaction?
- 3.5 Which is the rate-determining step of this reaction (Step1, step2 or step3)?
- 3.6 Calculate ΔH for the overall reaction?
- 3.7 Is the overall reaction endothermic or exothermic?

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4. Consider the following reaction: $A \rightarrow B + C$. (10 marks)



- 4.1 What is the order of reaction?
- 4.2 Calculate the rate constant (k) and give its units?
- 4.3 Determine the rate law for this reaction?
- 4.4 Calculate the half-life for this reaction?

) and give its units?
 is reaction?
 reaction?

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Dr. Jarukorn Sripradite

Important information

$$k = A \cdot \exp(-E_a/RT)$$

$$R = 8.314 \text{ J/K}\cdot\text{mol}$$

$$\ln(k_2/k_1) = -E_a/R \left((1/T_2) - (1/T_1) \right)$$

$$\log K_1 - \log K_2 = \frac{E_a}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

$$\ln k = -\frac{E_a}{R} \frac{1}{T} + \ln A$$

	Concentration-Time
Rate Law	Equation

$$\text{rate} = k[A]$$

$$\ln[A] = \ln[A]_0 - kt$$

$$\text{rate} = k$$

$$[A] = [A]_0 - kt$$

$$\text{rate} = k[A]^2$$

$$\frac{1}{[A]} = \frac{1}{[A]_0} + kt$$

$$\ln 1 = 0$$

$$e^1 = 2.71$$

$$\ln 2 = 0.693$$

$$e^2 = 7.39$$

$$\ln 3 = 1.098$$

$$e^3 = 20.09$$

$$\ln 4 = 1.38$$

$$e^4 = 54.60$$